CO-1
AT-3

AI

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1. A smart home automation system is being designed to manage Lighting, temperature, and security based on user behaviou and environmental conditions. The system must respond to real-time inputs such as motion detection. In this context, explain the different types of intelligent agents.

A) Intelligent agents in a smart home:

Simple-reflex: uses condition action rules on the current percent only.

ex: if motion detection and it is night, turn on the Light. It is fast and cheap but has nomemory, so it can't tell whether, a room is empty or the user is just sitting still.

model based reflex:

Keep an internal state of home.

ex: It refers the user is still in the Living room despite nomotion for 5min.

Goal based reflex:

choose actions to reach a goal such as "room at 24°c by 7pm" or house secured by 11pm. It plans ahead for example precooling the house before user returns it is flexible but does not weight trade-offs.

utility - based:

Maximizes a utility function that balances confort, energy cost and security.

ex:-

when comfort and electricity cost conflid, it picks the AC setting with the best overall score, It also copes with uncertainity.

Best choice:

a hybrid, with a Learning component o reflex rules for immediate safety and security, where speed matter.

- A model based Layer for tradeck the home's state.
- Goal based planning for scheduled routines.
- A utillity - based layer for trade-offs between comfort and energy.
- A Learing element that updates user perferences. over time.

This combination gives fast reactions content awareness and efficiency without sacrificing comfort.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. explain how uninformed search strategies like uniform cost search and Iterative deepening search (IDS) can help solve this problem. discuss the advantages and disadvantages of those algorithms in terms of intelligent agent.

A) uniform cost search (UCS)

expands the node with the lowest path cost (g(n)) using a priority queue. In amaze with varying step costs, it finds the cheapest path. It is complete and optimal if cost are possitive.

Iterative deepening search (IDS):
Runs depth-limited DFS with limits 0, 1, 2 and soon. It is complete, optimal for uniform step costs and uses little memory. It suits an unknown more because the goal depth is not known.

ues plos and cons:

Handles variable costs, but its memory use is exponential and it explores in the directions.

IDPS pros and cons:

Very memory efficient, but it re-expands upper levels (a small overhead, since most nodes are at the deepest level) and ignores step costs.

Link to agent types:

- A simple reflex agent (eg "turn left if wall ahead") can loop forever in a maze)
- A model based agent remembers visited cells and builts a map.
- A goal based uses search to plan a path to the exist.
- A utility based agent chooses among paths using time, energy and risk.

Agent design decides what the robot remembers and how it chooses between actions, so the search strategy only works well if the agent has a model of the maze.